

☐ Traffic Accident Severity Prediction: Tackling Class Imbalance

This repository contains a machine learning pipeline designed to predict the severity of traffic accidents based on environmental conditions. The project explores the challenges of **severe class imbalance** in real-world accident data and demonstrates how to build models that correctly identify minority classes.

☐ Objective

The objective is to move beyond the deceptive metric of overall "accuracy" when dealing with highly skewed datasets. By comparing a baseline model against implementations using **SMOTE** and **Class Weights**, this project aims to accurately predict severe and minor accidents that are otherwise masked by the overwhelming majority of standard accidents.

✂ Tech Stack

- **Python 3.x**
 - **Pandas & NumPy** (Data manipulation)
 - **Scikit-Learn** (Logistic Regression, scaling, evaluation)
 - **XGBoost** (Gradient-boosted decision trees)
 - **TensorFlow & Keras** (Artificial Neural Networks)
 - **Imbalanced-learn** (SMOTE resampling)
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☐ Dataset

The analysis utilizes a subset of traffic accident data, focusing specifically on South Carolina (SC) to test the models on regional environmental factors.

- **Target Variable:** severity (Scale 1-4)
 - **Key Features:** temperature_f and visibility_mi
 - **Data File:** data/sc_accidents.csv
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📈 Analysis & Visualizations

1. Feature Importance and Model Coefficients

This visualization breaks down the internal logic of the balanced Logistic Regression model,

highlighting how temperature and visibility influence accident severity.

2. Actual vs. Predicted Severity

This plot tracks the true accident severity against the model's predictions over a sample set, proving that the resampling techniques successfully mapped the minority classes.

□ Conclusion Summary

Relying on overall accuracy is a flawed approach for this dataset. While baseline models achieved ~87% accuracy, they failed to identify minority classes. Implementing SMOTE and adjusting class weights forced the models to learn the actual underlying patterns.

For a detailed breakdown of the findings, please see [conclusion.md](#).

□ Final Notes

This project demonstrates how misleading high accuracy can be when working with heavily imbalanced real-world datasets. By applying techniques such as SMOTE and class weighting, the models were able to better identify minority severity classes instead of favoring the dominant class. Future improvements can include richer traffic, weather, and road-condition features to build a more reliable and robust accident severity prediction system.

□ How to Run this Project

1. **Clone the repository:** git clone <https://github.com/suchismitab0511/Traffic-Accident-Severity-Prediction>
2. **Install dependencies:** pip install -r requirements.txt
3. **Run the notebook:** Open `accident_pred.ipynb` in Jupyter Notebook or Google Colab and run all cells.